SESSION 1 OF 2 · DIY REPAIR GUIDE

2018 Honda CR-V
1.5L Turbo (L15B7)

Three jobs, one coolant drain & front-end teardown. Read for a phone in the driveway — one step per screen.

- [ ] **Job 1 — A/C compressor + condenser** (system ends open/empty; recharge at shop)
- [ ] **Job 2 — ECT Sensor 2** in the lower radiator tank
- [ ] **Job 3 — Bar's Leaks HG-1** head-gasket stop-leak (½ bottle, cold)

⚠️ **THIS SESSION ENDS WITH THE A/C SYSTEM OPEN**

Refrigerant was professionally recovered. You will drive to a shop afterward to evacuate & recharge. Do not turn the A/C on until the shop is done.

HOW TO READ THIS GUIDE

Warning legend & the 5 rules

🛑 **STOP / DANGER** — do not skip. Safety or system-killer.

⚠️ **CRITICAL** — the step people get wrong. Slow down.

💡 **TIP** — makes the job easier or cleaner.

☑️ **Checkbox** — tap to tick off (or pen on paper).

🛑 **THE 5 RULES THAT SAVE THIS JOB**

1. **Never open A/C lines under pressure.**
2. **Cap every open line instantly — moisture is the enemy.**
3. **Fresh, oiled O-ring at every A/C fitting.**
4. **Hand-rotate the new compressor 8–10 turns before the drive.**
5. **HG-1 only in a cold engine, ½-bottle dose.**

PHASE 0 · PREP & SAFETY

Cold engine. Confirm. Stage parts.

🛑 **ENGINE MUST BE STONE COLD**

Cooling system is pressurized when hot. No draining, no cap off, no HG-1 until you can hold the radiator hose comfortably.

- [ ] **Refrigerant recovered — gauge reads 0 psi** (confirm with the shop that recovered it)
- [ ] **Battery NEGATIVE (−) terminal disconnected**, cable tucked aside
- [ ] Two **drain pans** ready (1 = coolant, 1 = old compressor oil)
- [ ] Parts staged: **compressor, condenser w/ drier, O-ring kit, PAG46 oil, coolant, ECT2 sensor + new O-ring, HG-1**

💡 **MEASURING CUP FOR OIL**

You'll drain and measure the old compressor oil — have a clear marked cup (mL) ready.

PHASE 1 · LIFT & DRAIN

Jack stands → splash shield → coolant

1. Loosen front wheel/lug area only if needed, raise the front and set on **jack stands** on solid ground.
2. Remove the **lower splash shield / undertray** (8/10mm bolts + push clips) to reach the radiator.
3. Place **drain pan #1** under the radiator's driver-side lower tank.
4. Open the **radiator petcock (drain plug)** by hand / flat screwdriver. Remove the cap or fill neck to speed flow.

🛑 **NEVER WORK UNDER A CAR ON A JACK ALONE**

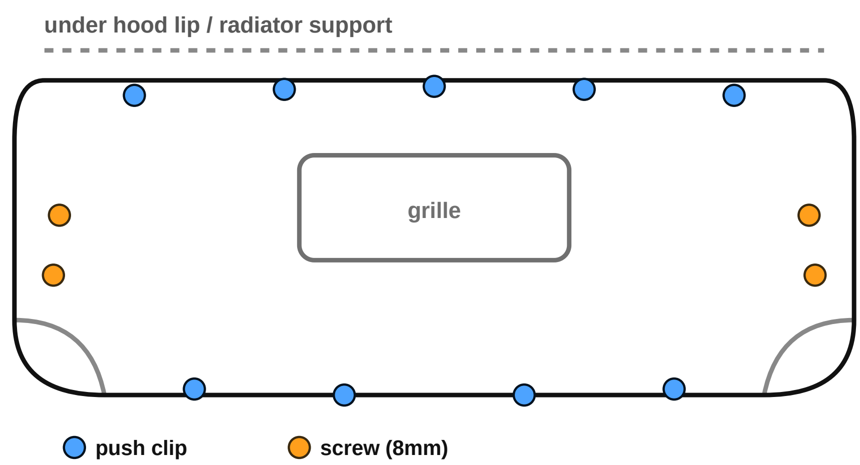
Jack stands rated for the weight, wheels chocked, level ground.

💡 **SAVE THE COOLANT VOLUME**

Note roughly how much drains — it tells you how much to put back. System holds ~5–6 qt.

PHASE 2 · FRONT BUMPER

Free the bumper cover for condenser room



Front bumper cover fastener map — top edge screws under the hood lip, wheel-well screws, and lower push-clips along the undertray line.

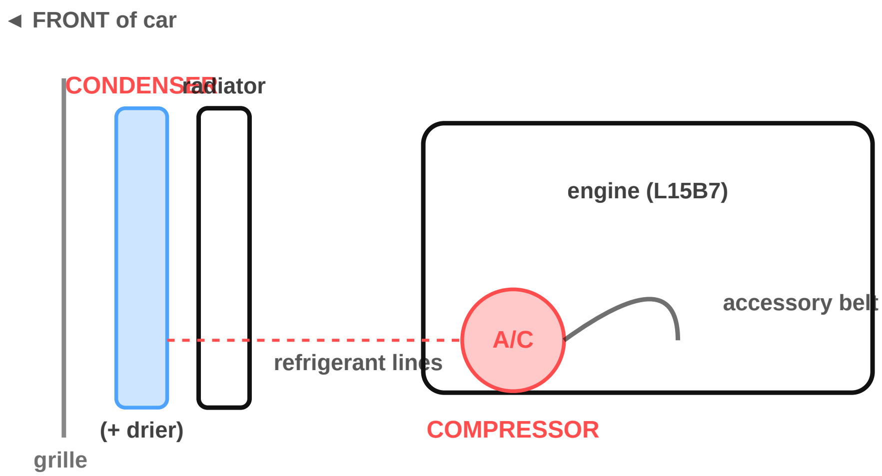
1. Open hood. Remove the **top-edge fasteners** along the radiator support (push clips / 8mm screws).
2. Turn wheels for access; remove **2–3 screws per wheel-well** where the liner meets the bumper.
3. Remove the **lower push-clips** along the bottom lip.
4. Release the side clips (pull the bumper corner outward at the fender) and lower the cover — support it or set it aside.

⚠️ **PLASTIC CLIPS BREAK — THAT'S NORMAL**

Have replacement clips on hand. Note which broke so you can match them.

PHASE 3 · A/C — STEP A

Open lines → CAP INSTANTLY



Front-end layout: condenser sits in front of the radiator; compressor is low on the front of the engine, driven by the accessory belt.

1. Unplug the **compressor clutch connector**.
2. Unbolt the **suction & discharge line flanges** at the compressor (single center bolt each).

NEVER OPEN A LINE STILL UNDER PRESSURE

Confirm 0 psi first (Phase 0). Refrigerant under pressure can cause frostbite/blindness.

CAP EVERY OPEN LINE IMMEDIATELY

The second a fitting is open, plug it. Moisture in an open system ruins the new parts and the recharge.

PHASE 3 · A/C — STEP B

Remove compressor · drain & MEASURE oil

1. Remove the **4 compressor mounting bolts** and lift the compressor out.
2. Tip the old compressor and **drain its oil into the measuring cup**. Record the amount in mL.
3. Rotate the old pulley/shaft while draining to get most of it out.

Why measure? The new compressor ships with a full oil charge. To avoid **overfilling** (which kills cooling), you set the new compressor to hold what you can account for across the whole system. **Final oil setting is done at the shop with the recharge — bring your measured number.**

⚠️ **WRITE THE NUMBER DOWN**

Old oil out = your reference. Give it to the shop so total system oil is correct.

PHASE 3 · A/C — STEP C

Remove the old condenser

1. Disconnect the **two refrigerant lines** at the condenser — **cap them instantly**.
2. Remove the condenser **upper mounting bolts / brackets**.
3. Lift the condenser **up and out** (it usually drops into lower rubber mounts).

⚠️ **MIND THE RADIATOR FINS**

Don't jam the condenser against the radiator on the way out — bent fins hurt cooling.

💡 **NEW CONDENSER INCLUDES THE DRIER**

Good — a fresh receiver/drier is exactly why we replace the condenser with an open system.

PHASE 3 · A/C — STEP D

Prep the new compressor

1. **Set the oil charge**: match new compressor oil to what the system needs (drain/re-measure per the new unit's instructions; final setting confirmed at the shop).

2. Pour a measured amount of **PAG46 into the SUCTION port**.

3. **HAND-ROTATE the pulley 8–10 full turns** to pre-lube the internals.

🛑 **HAND-ROTATE BEFORE IT EVER RUNS**

Skipping this can seize a dry compressor on first start. 8–10 slow turns, by hand, no exceptions.

⚠️ **CHECK THE REFRIGERANT TYPE FIRST**

The 2018 1.5T very likely uses R-1234yf, not R-134a. Generic PAG46 is the R-134a oil — an R-1234yf system may need Honda's own oil (ND-OIL 12 / POE). Read the underhood A/C label and confirm the correct oil for your VIN before adding any.

Oil: use the Honda-spec oil for your refrigerant (PAG46 only if confirmed). **Exact charge + final oil balance are set at the shop with the recharge — bring your measured old-oil number.**

PHASE 3 · A/C — STEP E

Add oil to condenser · install · torque

1. Add **~1 oz (≈30 mL) PAG46** to the new condenser before fitting.
2. Install condenser, then compressor, into their mounts. Torque the **4 compressor bolts**.
3. At **every** fitting, install a **NEW, PAG-oiled O-ring**. Reconnect line flanges.
4. Torque the line flange bolts. Reconnect the clutch connector.

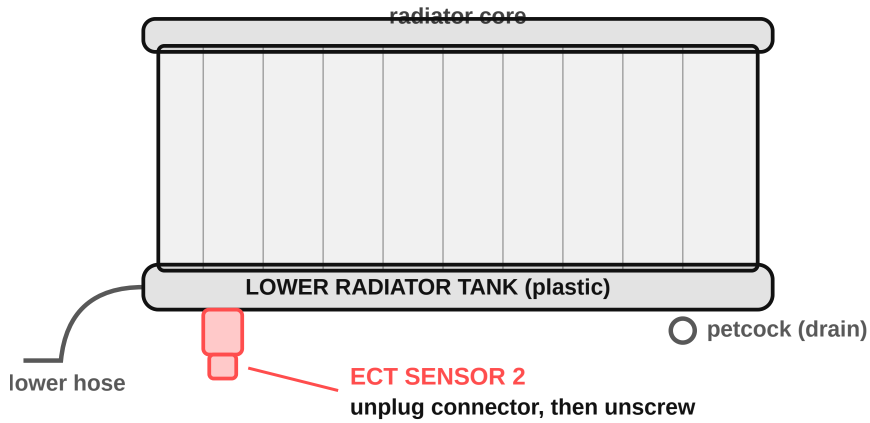
Torque — line flange bolts: ≈9.8–12 N·m (7–9 lb-ft / 87–106 in-lb) · **Compressor mount bolts:** ≈24–25 N·m (18 lb-ft) · **Condenser line fittings:** ≈9.8 N·m (7 lb-ft) ***Typical Honda A/C values — verify against the 2017–2022 CR-V service manual for your VIN before final torque.**

A MISSED O-RING IS THE #1 LEAK

Every fitting. New. Oiled. No reused O-rings, ever.

PHASE 4 · ECT SENSOR 2

Swap the lower-radiator ECT2



ECT Sensor 2 threads into the lower radiator tank (driver side, near the lower hose). Coolant is already drained from Phase 1.

1. **Unplug** the ECT2 connector.
2. Unscrew the old sensor with a **deep socket — measure the flats first** (reports split between 17 mm and 19 mm; bring both).
3. Fit the **new O-ring** to the new sensor; thread in by hand, then snug.

⚠️ **DO NOT OVERTIGHTEN INTO THE PLASTIC TANK**

It threads into a plastic radiator tank. No metal-boss torque spec applies — seat by hand on the new O-ring, then hand-tight + a small nudge only. Overtightening cracks the tank and causes a coolant leak.

PHASE 5 · REASSEMBLE

Bumper → splash shield → battery

- [ ] Reinstall **bumper cover — replace every broken clip** with new ones
- [ ] Reconnect wheel-well screws, top-edge fasteners, lower clips
- [ ] Reinstall the **lower splash shield / undertray**
- [ ] Confirm the radiator **petcock is CLOSED** before refilling
- [ ] **Reconnect the battery (−)** terminal, snug it

⚠️ **PETCOCK CLOSED?**

Easy to forget — a fresh fill will pour straight onto the floor if it's open.

PHASE 6 · COOLANT + HG-1 + BLEED

Refill 50/50 · add HG-1 cold · burp the air

1. With engine **COLD**, slowly fill through the **expansion / degas tank** with 50/50 **Honda Type 2 (blue) coolant**. This engine has **NO bleed bolt** — it self-bleeds through the tank. Install the cap **loosely**.
2. Add ½ **bottle of Bar's Leaks HG-1** now, while cold (system holds ~6.2 L ≈ 6.5 qt — ½-bottle bracket).
3. Start the engine. Air purges automatically back to the degas tank as it warms.
4. Set **heater to HIGH, fan HIGH**. Idle ~**15 min**, watching the temp gauge, until the radiator fan cycles **twice**.
5. Shut off, let cool, then **top the expansion tank to MAX**. Re-check and top off over the next few drive cycles as trapped air works out.

🛑 **HG-1 GOES IN COLD, ½ BOTTLE ONLY**

Never pour stop-leak into a hot/running-hot engine. Half bottle for this ~5–6 qt system.

⚠️ **WATCH THAT TEMP GAUGE**

If it climbs toward hot or the upper hose stays cold, you have an air pocket — stop, let cool, re-burp. Do not overheat.

PHASE 7 · VERIFY & LEAK CHECK

Final checks before it moves

- [ ] **Every A/C O-ring is new & oiled** (the #1 leak source)
- [ ] All A/C **fittings torqued** (flanges + compressor bolts)
- [ ] You **hand-rotated the compressor** 8–10 turns
- [ ] No **coolant leaks** at ECT2, petcock, hoses after warm idle
- [ ] Coolant topped off; reserve at MAX; cap tight
- [ ] Old compressor oil amount **written down** for the shop

DO NOT TURN THE A/C ON

The system is open/empty. Running it dry destroys the new compressor. Straight to the shop.

BEFORE YOU DRIVE

Final drive-off checklist

- [ ] Battery reconnected, tools out of the engine bay
- [ ] Car off the stands, wheels torqued if removed, on the ground
- [ ] Splash shield & bumper fully fastened
- [ ] Coolant full, no leaks, temp normal at idle
- [ ] **A/C left OFF** — no exceptions
- [ ] Measured old-oil note + parts receipts in hand for the shop

💡 **NEXT: THE SHOP**

They evacuate the system, verify no leaks under vacuum, set the final oil charge, and recharge the refrigerant. Then — and only then — the A/C runs.

This is Session 1 of 2. Session 2 covers the follow-up work in your two-session plan.

REFERENCE

What happens at the shop

1. **Evacuate** the open system under deep vacuum (pulls out air & moisture).
2. **Vacuum leak test** — holds vacuum = your O-rings sealed.
3. **Set total oil charge** (PAG46) using your measured old-oil number.
4. **Recharge** refrigerant to Honda spec, then verify vent temps.

Bring to the shop: the measured old compressor oil amount (mL), the new parts' oil-spec sheets, and a note that a fresh condenser+drier were installed.

Guide generated for a 2018 Honda CR-V 1.5T (L15B7). Verify every torque/spec against the factory service manual for your VIN before final assembly.